

PERSONAL INFORMATION

Muhammad Hamad Khaliq

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Sex Male | Date of birth 01 Jan 2003 | Nationality Pakistani



WORK EXPERIENCE

OCT 2024 – MAY 2025 **Python Developer:** Devticks, Bahawalpur, Pakistan

- Developed and integrated RESTful APIs to enable seamless communication between backend services and client applications.
- Implemented unit testing and automated testing frameworks using Pytest and Unittest to ensure code quality and system reliability.
- Collaborated with cross-functional teams to debug, optimize, and deploy efficient backend solutions.

JUN 2024 – SEP 2024

Software Engineer: Tixel, Lahore, Pakistan

- Development: Designed and maintained scalable web and backend applications using Python, Django, and RESTful APIs.
- DevOps: Worked with CI/CD pipelines, version control (Git), and deployment automation for faster software delivery.
- Cybersecurity: Followed secure coding practices and API authentication mechanisms (JWT, OAuth) to protect user data.
- Collaboration: Worked in Agile/Scrum environment and contributing to sprint planning

JUN 2023 – DEC 2023

Web Developer: Infinite IT Solution, Bahawalpur, Pakistan

- Developed responsive web pages and UI components using HTML, CSS, JavaScript, and improved overall website performance.
- Collaborated with designers and backend developers to implement features, fix bugs, and enhance user experience.
- Worked with Git/GitHub, integrated APIs, and supported end-to-end development across multiple client projects.

EDUCATION

SEP 2021 – JUN 2025

Institution

Major Subjects

Bachelor of Science [BS (16 Years of Education)]

Islamia University of Bahawalpur, Pakistan

3.69 /4.0 CGPA

Artificial Intelligence • Machine Learning • Deep Learning / Artificial Neural Networks • Natural Language Processing (NLP) • Computer Vision • Reinforcement Learning • Knowledge Representation & Reasoning • Analysis of Algorithms • Web Programming / Mobile Application Development

JUN 2019 – MAY 2021

Institution

Major Subjects

Higher Secondary School Certificate [Science Group]

Government Degree College, Kehror Pacca, Pakistan

B GRADE

Biology • Physics • Chemistry

AUG 2017 – March 2019

Institution

Higher Secondary School Certificate [Science Group]

Al-Ghazali Public High School, Dunyapur Lodhran, Pakistan

B GRADE

PROJECTS

- **Advance Laser Interception System in Islamia University (FYP)**
Engineered an Arduino-based automated laser defense system capable of real-time detection, tracking, and interception of moving targets with high precision. Integrated ultrasonic sensors, servo motors, and laser modules to achieve dynamic object recognition and rapid response accuracy. Programmed efficient embedded logic in C/C++ (Arduino IDE) to perform intelligent decision-making and automated target engagement.
 - **Dectron – Autonomous AI Walking Robot**
Developed a compact autonomous robot integrating multi-model AI systems for locomotion and object detection. Designed walking mechanics using reinforcement learning-based motion control, PID tuning, and sensor fusion from ultrasonic, IR, and IMU modules. Implemented real-time object detection through YOLO/OpenCV, enabling obstacle avoidance and stable navigation. Optimized the control pipeline for low-latency inference on embedded hardware.
 - **Mental Health Care – AI Chatbot**
Engineered an AI-driven mental health support system using NLP pipelines, transformer-based language models, and intent-classification architectures to process user queries and generate context-aware responses. Implemented sentiment analysis, fallback handling, and optimized dialogue flow using Python, TensorFlow/PyTorch, and Rasa/DialogFlow.
 - **Stock Price Prediction using LSTM**
Developed a time-series prediction model using Long Short-Term Memory (LSTM) networks to forecast stock prices. Collected and preprocessed historical stock data using Pandas and NumPy for trend analysis. Achieved accurate predictions through model tuning and visualization with Matplotlib and Seaborn.
 - **Hand Gesture Recognition using CNN**
Built a Convolutional Neural Network (CNN) model to classify real-time hand gestures from image datasets. Utilized OpenCV for image preprocessing and data augmentation to enhance model accuracy. Implemented and trained the model in TensorFlow/Keras, achieving high recognition precision for gesture-based control systems.
 - **Deepfake Detection System**
Developed a CNN-based model to detect and classify deepfake videos and manipulated facial images with high accuracy. Utilized Convolutional Neural Networks (CNNs) and Transfer Learning (VGG16/ResNet) to extract visual and temporal features. Implemented preprocessing using OpenCV and trained the model in TensorFlow/Keras for robust deepfake identification.
 - **2D Machine (Convert Digital Art to Paper Art)**
Designed and developed a custom 2D printer capable of converting digital art into physical drawings on paper using stepper motor control. Utilized Arduino microcontroller with servo and stepper motors to achieve precise 2-axis motion and pen control. Programmed motion algorithms in C/C++ (Arduino IDE) for accurate path plotting and real-time printing execution.
 - **Intelligent Traffic Management System**
Built a real-time traffic optimization system using computer vision for vehicle detection and reinforcement learning (DQN) to dynamically adjust traffic light timings. Used YOLOv8/OpenCV for detecting congestion density and fed environment states into a learning agent to minimize waiting time and reduce traffic jams.
 - **Multi-Agent Research Assistant Chatbot**
Built an advanced research assistant chatbot using LLMs, Retrieval-Augmented Generation, and vector search (FAISS/Pinecone) for academic document summarization and query analysis. Implemented multi-agent workflows for semantic search, citation extraction, and tool-calling, with a FastAPI backend and LangChain-based orchestration.
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ADDITIONAL INFORMATION

Conferences

AI Project Representative: AI Expo 2025, The Islamia University of Bahawalpur (IUB).

Python Instructor: ACM Student Chapter-2 Workshop, IUB (2024).

AI Project Representative: AI Expo 2023, The Islamia University of Bahawalpur (IUB).

Participant: GDSC Web Development Workshop, IUB (2022).

Conference Attendee: First International Conference on Machine Perception & AI (ICMPAI-22), IUB (June 2022).

Awards

Position – AI Expo 2023 (IUB)

Project: 2D Machine – Convert Digital Art to Paper Art

Awarded 1st place at the AI Expo held at The Islamia University of Bahawalpur for developing an innovative system that transforms digital artwork into precise paper-based output.

Certificates

TensorFlow Developer Certificate – TensorFlow (Google)

Validated practical skills in building, training, and deploying deep-learning models using TensorFlow.

• Google Professional Machine Learning Engineer – Google Cloud

Demonstrated expertise in designing, automating, and deploying ML models at scale.

• Microsoft Certified: Azure AI Engineer Associate – Microsoft

Focused on building, evaluating, and deploying AI solutions using Azure Cognitive Services.

• AWS Certified Machine Learning – Specialty – Amazon Web Services

Advanced knowledge of cloud-based machine learning workflows and scalable ML pipelines.

• Google Professional Cloud Developer – Google Cloud

Skilled in developing, testing, and deploying cloud-native applications on GCP.

• Certified ScrumMaster (CSM) – Scrum Alliance

Certified understanding of Agile principles and effective Scrum team collaboration.

• Certified Secure Software Lifecycle Professional (CSSLP) – (ISC)²

Demonstrated ability to integrate security best practices into every phase of software development.

• GitHub Foundations / GitHub Actions – GitHub

Validated competency in Git, version control, GitHub CI/CD, and workflow automation.

• Microsoft Certified: Azure DevOps Engineer Expert – Microsoft

Expertise in DevOps pipelines, CI/CD, IaC, and cloud deployments.

• Elements of AI – University of Helsinki

Completed globally recognized AI fundamentals course focused on modern AI concepts.

SKILLS

Programming Languages: Python | C++ | JavaScript | HTML

AI / Machine Learning: Machine Learning | Deep Learning | Neural Networks | Reinforcement Learning | Generative AI

Frameworks & Libraries: TensorFlow | PyTorch | Scikit-Learn | Keras | OpenCV | Pandas | NumPy

Robotics & Embedded: Arduino | Raspberry Pi | Microcontrollers | Sensor Integration | Embedded AI

Tools & Platforms: Jupyter | NoteBook | VSCode | Google Colab | Linux | Anaconda

REFERENCES

Name: Dr. Najia Saher
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Organization: Dept. of Artificial Intelligence
Islamia University of Bahawalpur,
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Name: Dr. Ghulam Gilanie
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